🛹 Energy Skate Park Simulation Worksheet

Topic: Energy Transformations
Level: National 5 Engineering Science
Simulation Link: https://phet.colorado.edu/sims/html/energy-skate-park/latest/energy-skate-park_all.html

🔍 Learning Objectives

- Understand the relationship between kinetic and potential energy.
- Observe energy conservation in a closed system.
- Investigate the effects of mass, friction, and track design on energy transfer.

🧠 Part 1: Getting Started

Instructions: Open the simulation and select the "Intro" tab.

1. Place the skater on the track. What happens when you release them from the top of the ramp?
   ✎ The skater follows the path of the ramp and travels downwards
2. Turn on the **Bar Graphs** and **Pie Chart**. What do these represent?
   ✎ These represent the total energy and also constituent energy types

   ✎ _____

   ✎ _____

⚙️ Part 2: Energy Observations

Instructions: Use the simulation to answer the following.

Position on Track	Kinetic Energy	Potential Energy	Total Energy
Top of Ramp	Zero (0%)	Max (100%)	Max (100%)
Middle of Ramp	Half (50%)	Half (50%)	Max (100%)
Bottom of Ramp	Max (100%)	Zero (0%)	Max (100%)

1. What do you notice about the **total energy** at each point?
   ✎ It does not change
2. How does the **skater's speed** relate to their kinetic energy?
   ✎ The faster the skater, the greater the kinetic energy

   ✎ _____

🧪 Part 3: Experimenting with Variables

Instructions: Try changing the skater's **mass** and **friction** settings.

1. Increase the skater's mass. What effect does this have on:
    - Potential energy? ✎ Potential energy increases with mass

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    - Kinetic energy? ✎ Kinetic energy increase with mass

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2. Add friction to the track. What happens to the total energy over time?
   ✎ Total energy remains constant. At rest there is zero potential energy and zero kinetic energy. All the energy is heat energy caused by friction in wheel bearings and wheel surfaces.

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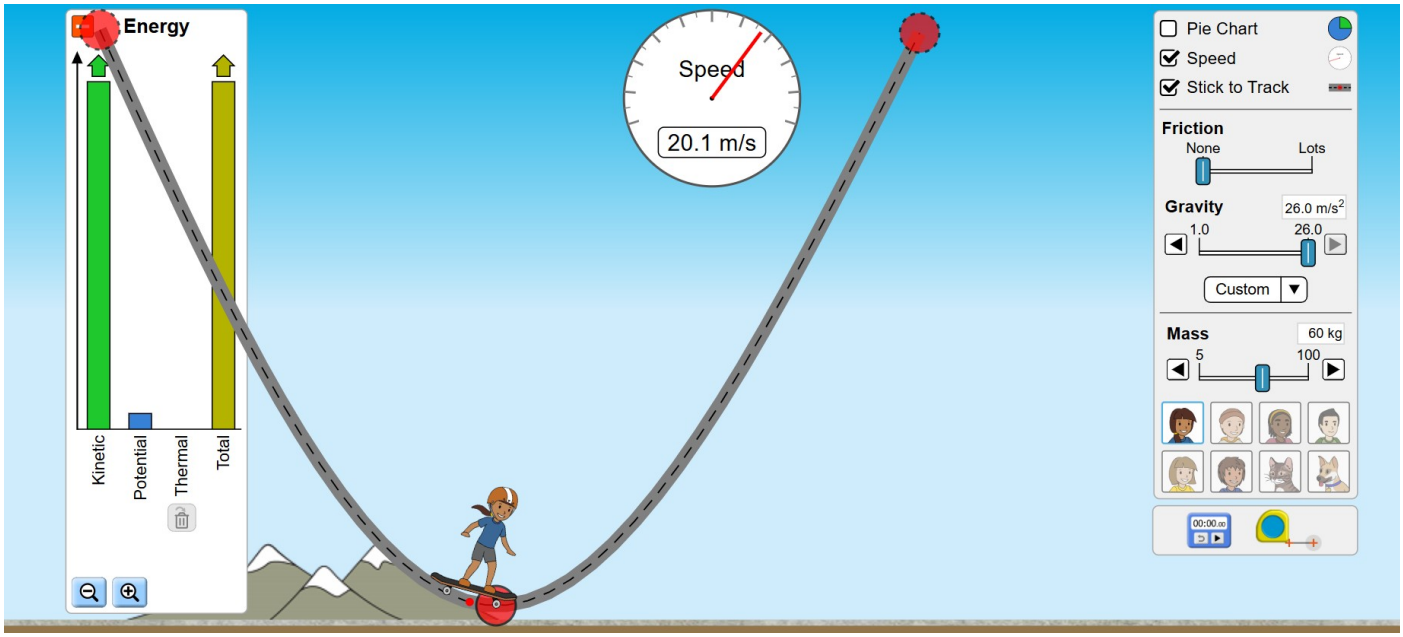
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🎨 Part 4: Design Challenge

Instructions: Switch to the "Playground" tab and design your own track.

1. Create a track where the skater reaches maximum speed without flying off.
    - Sketch your track below or describe its shape:
      ✎ The track needs to have the steepest angle and greatest height to achieve maximum speed. To avoid flying off, the shape of the track after the bend needs to be identical to the downward slope (assuming no energy losses)

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2. What design features helped maintain energy efficiency?
   ✎ Zero friction (no heat energy losses)

🧠 Reflection

1. In your own words, explain how energy is conserved in the simulation.
   ✎ Energy is conserved in a system because it cannot be created or destroyed, but only transferred between different forms or transformed from one type to another. In a closed system, the total amount of energy remains constant over time; the energy before any process equals the total energy after, even if it changes shape, such as from potential energy to kinetic energy
2. Why is this concept important in real-world engineering applications?
   ✎ allows engineers to design more efficient and sustainable systems by minimizing energy losses and optimizing energy transformations. This fundamental law guides engineers in developing innovative solutions across various fields, from transportation and electronics to renewable energy and building design, ultimately reducing waste, lowering costs, and promoting responsible resource use.

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